

THE BUBBLES IN A DROP

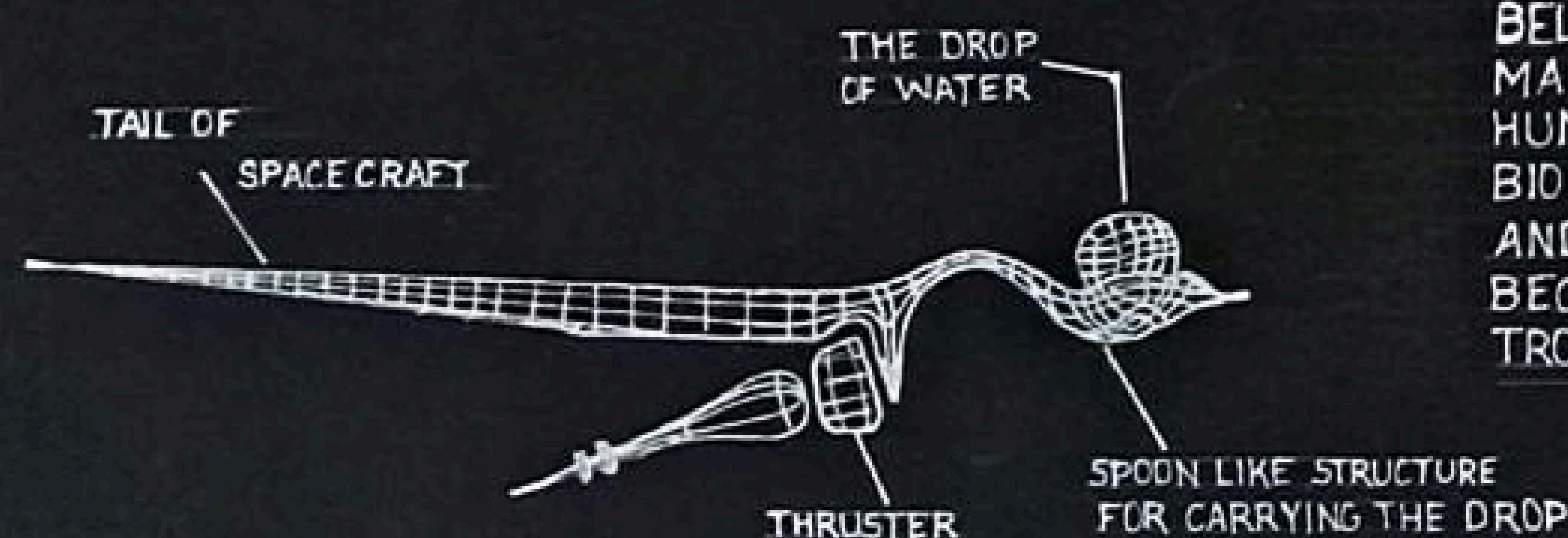
PROJECT SUMMARY

THE PROPOSAL, "THE BUBBLES IN A DROP", EXPLORES THE IDEA OF LIFE ABOARD A MULTIGENERATIONAL SPACECRAFT. MANY GENERATIONS AFTER HUMANITY HAS LEFT EARTH THE PROJECT FOCUSES ON THE FUNDAMENTAL IMPORTANCE OF WATER AS THE ESSENCE OF LIFE, WHILE EXPLORING THE VERY EXISTENCE OF LIFE ITSELF. IT CHALLENGES CONVENTIONAL PHILOSOPHIES OF PURPOSE AND BELONGING, ENCOURAGING REFLECTION ON HUMANITY'S PLACE IN THE UNIVERSE AND THE MEANING OF HOME WHEN EARTH EXISTS ONLY AS A DISTANT MEMORY.

INTRODUCTION

THERE CAME A TIME WHEN ASTRONOMERS DISCOVERED THAT THE MILKY WAY WOULD BECOME UNINHABITABLE WITHIN A FEW CENTURIES. FORCING HUMANITY TO SEARCH FOR A NEW HOUSE BEYOND EARTH IN RESPONSE, HUMANKIND EMBARKED ON ITS GREATEST ENGINEERING ACHIEVEMENT—A MULTIGENERATIONAL SPACECRAFT DESIGNED TO SUSTAIN LIFE ACROSS COUNTLESS GENERATIONS DURING AN INTERSTELLAR JOURNEY.

THE SPACECRAFT'S FORM DRAWS INSPIRATION FROM A SPOON CARRYING A SINGLE DROP OF WATER. WATER, THE MOST FUNDAMENTAL REQUIREMENT FOR LIFE, BECAME THE VESSEL'S PRIMARY CARGO AND THE FOUNDATION OF ITS ECOSYSTEM. THE DROP, APPROXIMATELY 300KM IN DIAMETER, CONTAINS A 25KM SOLID CORE OF FERTILE SOIL, GENETIC ARCHIVES, AND RARE ELEMENTS ESSENTIAL FOR THE CONTINUATION OF LIFE. SURROUNDING THIS CORE LIES A VAST BODY OF PURIFIED WATER, CREATING A SELF-SUSTAINING AQUATIC ENVIRONMENT PROTECTED FROM THE HARSH CONDITIONS OF SPACE.



BELIEVING THAT AQUATIC ECOSYSTEMS ARE EASIER TO MAINTAIN AND EXPAND WITHIN A CLOSED-LOOP SYSTEM, HUMANITY CHOSE TO PRESERVE MUCH OF EARTH'S BIODIVERSITY IN WATER. AQUATIC ORGANISMS, MICROORGANISMS, AND CAREFULLY SELECTED BIOLOGICAL RESOURCES BECAME THE BASIS OF A NEW LIVING WORLD TRAVELLING THROUGH THE COSMOS.

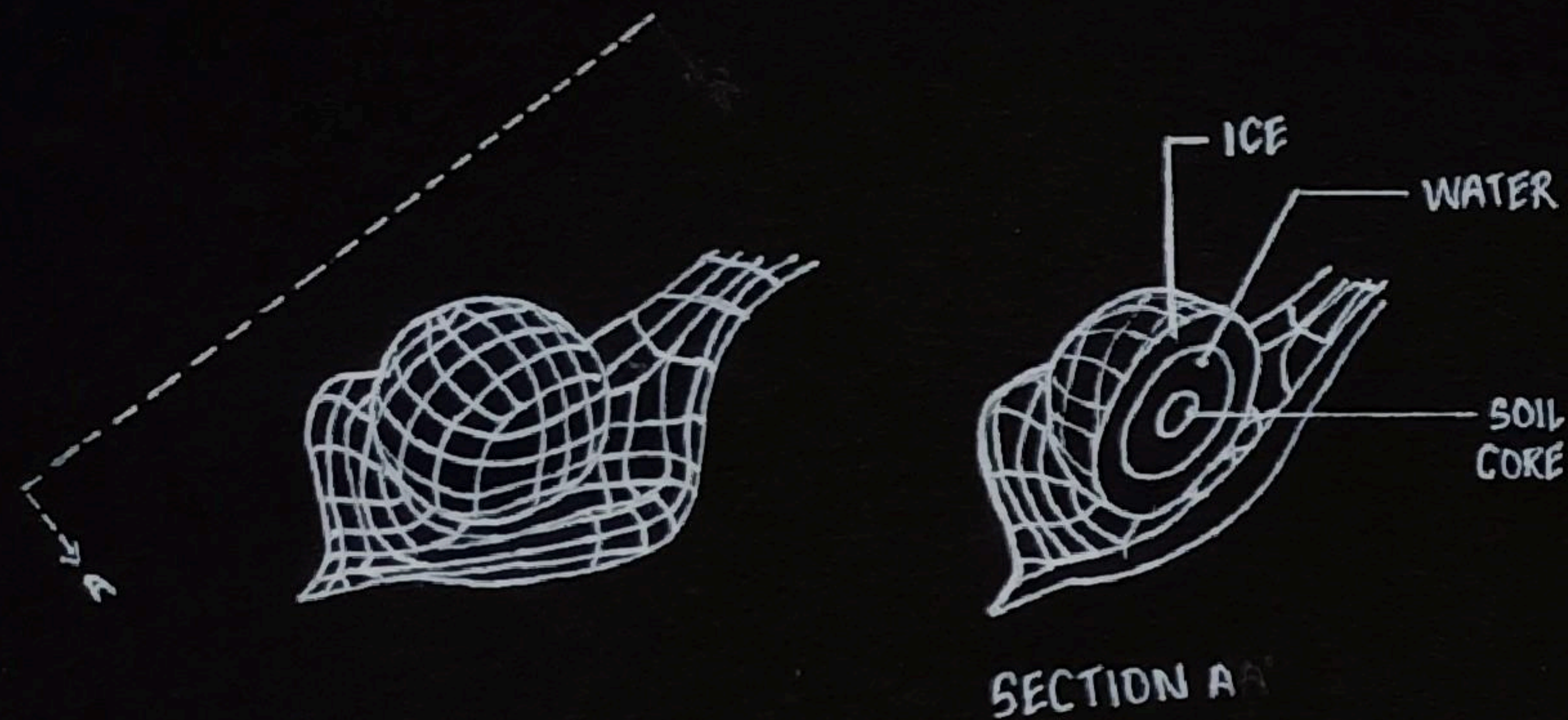
TO ENSURE THE SURVIVAL OF CIVILISATION HUMANITY DIVIDED ITSELF INTO TWO POPULATIONS. A SMALL NUMBER OF INDIVIDUALS INHABIT THE ENLONGATED TAIL STRUCTURE, RESPONSIBLE FOR MONITORING, MAINTAINING, AND GUIDING THE VESSEL. THE MAJORITY RESIDE WITHIN THE WATER-FILLED DROP, WHERE GENERATIONS OF GENETIC ADAPTATION GRADUALLY TRANSFORMED HUMAN LIFE ENHANCED FOR LONGER LIFESPANS AND GREATER COMPATIBILITY WITH AQUATIC ENVIRONMENTS. THESE DESCENDANTS EVOLVED TO LIVE DEEP WITHIN THE WATER ITSELF.

OVER CENTURIES, WATER BECAME MORE THAN A RESOURCE—IT BECAME THEIR ATMOSPHERE, THEIR LANDSCAPE, AND THEIR HOME. SHIELDED FROM HARMFUL COSMIC RADIATION, SUCCESSIVE GENERATIONS ADAPTED TO A WAY OF LIFE UNIMAGINABLE TO THEIR ANCESTORS. FOR THEM, EARTH EXISTS ONLY AS A DISTANT MEMORY PRESERVED IN STORIES, WHILE THE DROP ITSELF HAS BECOME AN ENTIRE WORLD.

DESIGN CONCEPT AND ASSUMPTION

SITE

OVER CENTURIES, THE OUTER LAYERS OF THE DROPLET HAVE FROZEN INTO ICE, WHILE LIQUID WATER REMAINS AROUND THE CORE, SUSTAINING A DIVERSE AQUATIC ECOSYSTEM AND AN ADAPTED HUMAN POPULATION.



LIVING SYSTEM OF THE UNDERWATER CIVILIZATION

GENERATIONS OF GENETIC ADAPTATION HAVE ENABLED HUMANS TO SURVIVE AND THRIVE UNDERWATER. THEIR DAILY LIVES REVOLVE AROUND HUNTING, GATHERING, AND NAVIGATING THE VAST AQUATIC WORLD SURROUNDING THE CORE. OXYGEN IS PRODUCED THROUGH THE ELECTROLYSIS OF WATER, WHILE THE EXTRACTED HYDROGEN POWERS THE SPACECRAFT'S SYSTEMS.

A SMALL POPULATION RESIDES WITHIN THE TAIL STRUCTURE OF THE VESSEL, RESPONSIBLE FOR OPERATING, MAINTAINING, AND GUIDING THE SPACECRAFT. SEPERATED FROM THE AQUATIC CIVILIZATION FOR CENTURIES, THESE INDIVIDUALS HAVE GRADUALLY PASSED FROM REALITY INTO LEGEND. TO THE INHABITANTS OF THE DROP, THEY ARE NO LONGER ENGINEERS OR CARETAKERS - THEY ARE KNOWN SIMPLY AS "THE GODS".

MEANING OF HOME

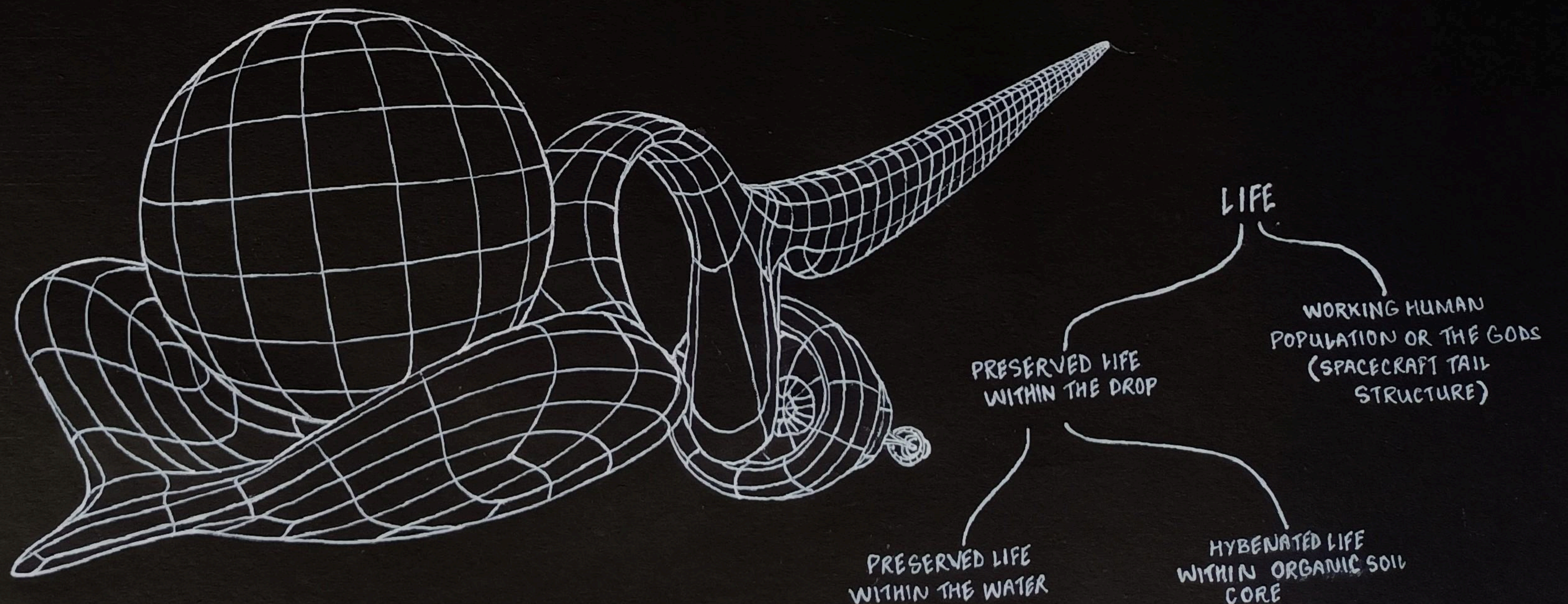
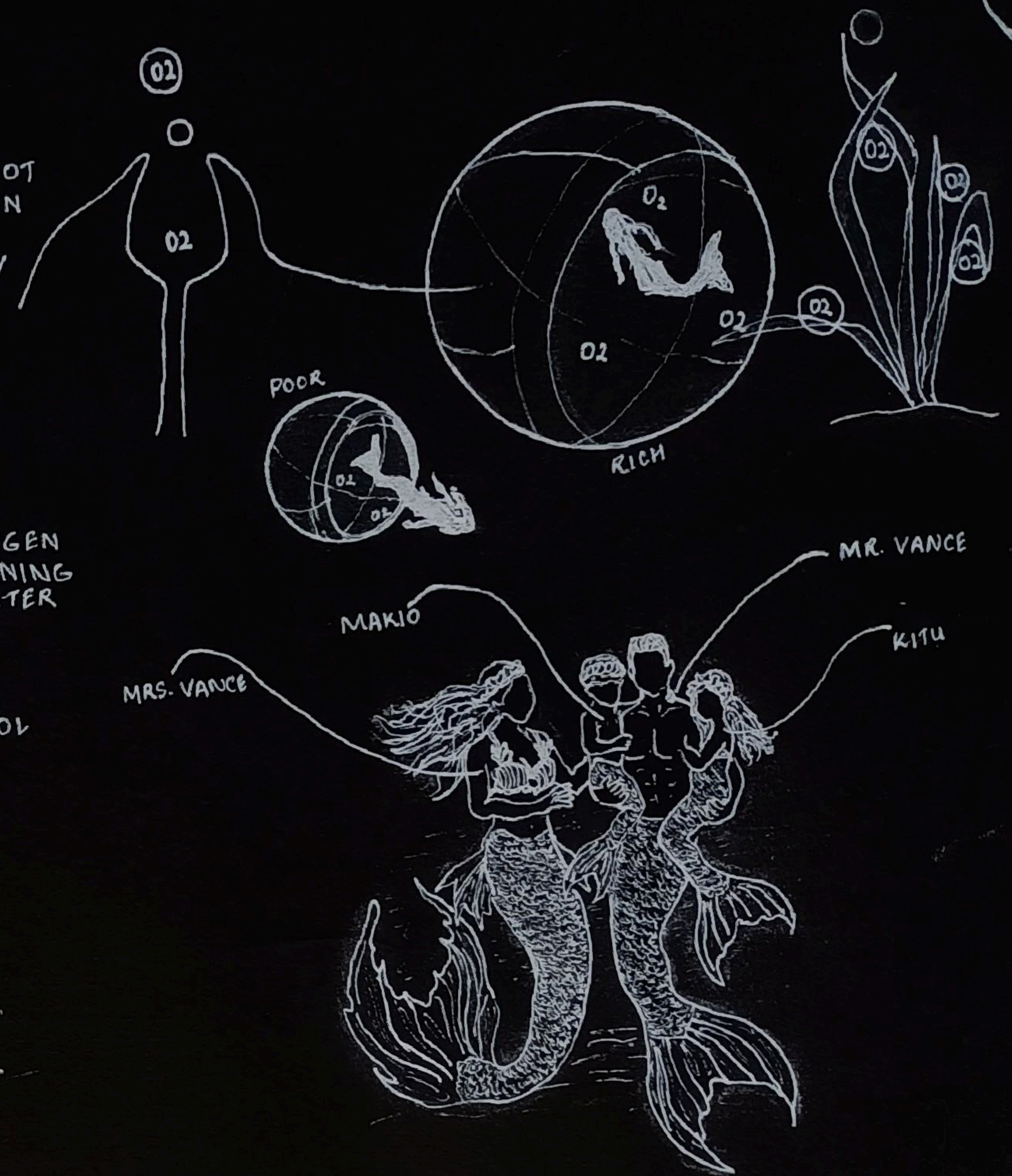
FOR THE INHABITANTS OF THE DROP, A HOME IS NOT DEFINED BY WALLS OR LAND. HOME IS AN OXYGEN BUBBLE - A PLACE TO BREATHE, REST, FIND PRIVACY, CONNECT WITH FAMILY AND TEMPORARILY ESCAPE THE PRESSURES OF THE SURROUNDING AQUATIC WORLD.

SOCIAL HIERARCHY

SOCIAL STATUS IS MEASURED BY ACCESS TO OXYGEN. INDIVIDUALS AND FAMILIES CAPABLE OF MAINTAINING LARGER AND MORE STABLE BUBBLES POSSESS GREATER WEALTH, INFLUENCE, AND SECURITY. THOSE WITH SMALLER BUBBLES LIVE CLOSER TO SURVIVAL CONDITIONS, CREATING A SOCIETY WHERE OXYGEN FUNCTIONS AS BOTH A RESOURCE AND A SYMBOL OF POWER.

FICTIONAL USER SCENARIO

THE PROJECT EXPLORES HABITATION THROUGH THE STORY OF THE FICTIONAL AQUA HUMAN FAMILY, WHOSE DAILY ACTIVITIES DEMONSTRATE THE USE, MAINTENANCE, AND GROWTH OF A BUBBLE HOUSE.



NEED WANT & DESIRE

NEED

AQUA HUMANS REQUIRE POCKETS OF OXYGEN TO BREATHE, RECOVER AND FIND RELIEF FROM THE SURROUNDING VISCOUS AQUATIC ENVIRONMENT THESE OXYGEN-FILLED SPACES SERVE AS SHELTERS, PLACES OF REST, AND LONG-TERM HABITATS, BECOMING ESSENTIAL FOR BOTH SURVIVAL AND DAILY

LIFE ANCHORED TENSILE MEMBRANES FORM A NAVIGATION NETWORK TROUGHOUT THE CAVITY, MAINTAINING SAFE CIRCULATION GAPS BETWEEN HABITATS THE SYSTEM GUIDES MOVEMENT, CONNECTS SETTLEMENTS, AND PREVENTS INHABITANTS FROM BECOMING TRAPPED WITHIN ISOLATED UNDERWATER SPACES.

WANT

A HABITAT THAT CAN GROW, ADAPT, AND MOVE WITH ITS INHABITANTS. RATHER THAN FIXED BUILDINGS, THE SOCIETY SEEKS FLEXIBLE HOMES THAT RESPOND TO CHANGING OXYGEN AVAILABILITY AND FAMILY NEEDS.

DESIRE

TO POSSESS LARGER AND MORE STABLE OXYGEN BUBBLES, SYMBOLISING COMFORT, SECURITY, AND SOCIAL STATUS. THE DESIRE FOR GROWTH DRIVES THE CONTINUOUS COLLECTION AND MERGING OF OXYGEN BUBBLES, ALLOWING HOMES TO EVOLVE OVER TIME.

SPATIAL PROGRAM

PRIVATE SPACES

PARENTS' ROOM [B1]
CHILDREN'S BEDROOM [B2]

COMMON SPACES

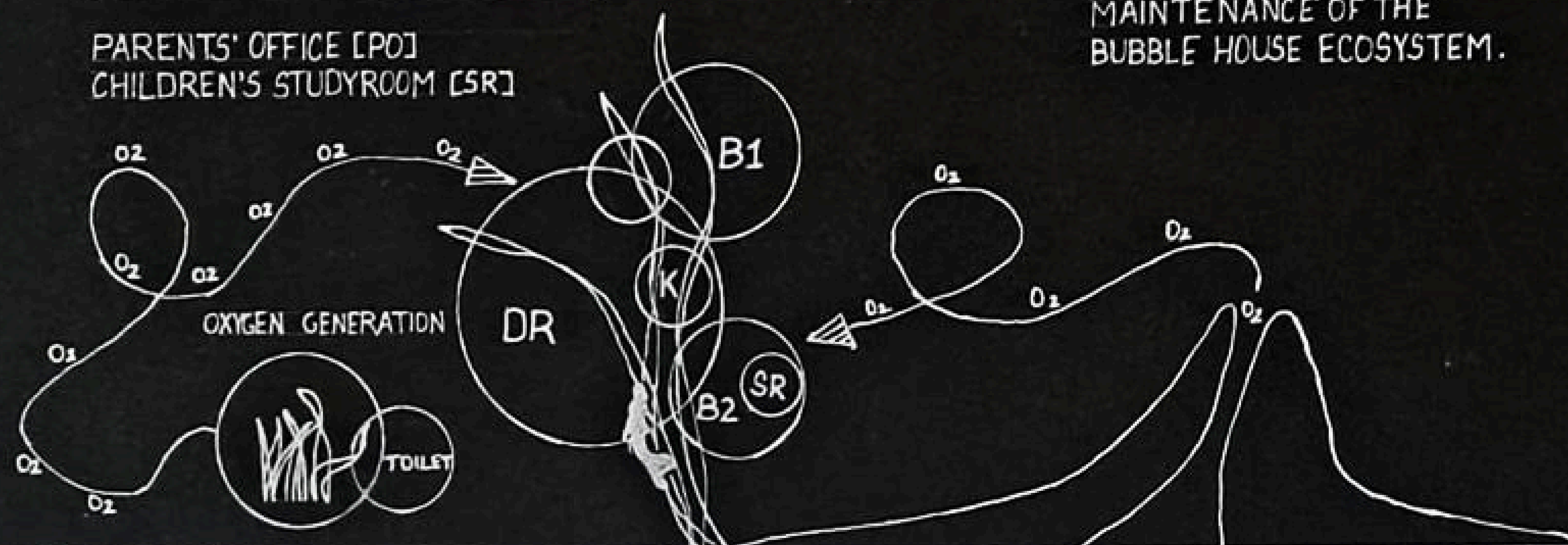
KITCHEN [K]
FAMILY GATHERING SPACE [DR]

ECOLOGICAL SPACE

OXYGEN GARDEN - A CULTIVATION ZONE WHERE AQUATIC PLANTS GENERATE OXYGEN AND SUPPORT THE GROWTH AND MAINTENANCE OF THE BUBBLE HOUSE ECOSYSTEM.

WORK AND LEARNING SPACES

PARENTS' OFFICE [PO]
CHILDREN'S STUDYROOM [SR]

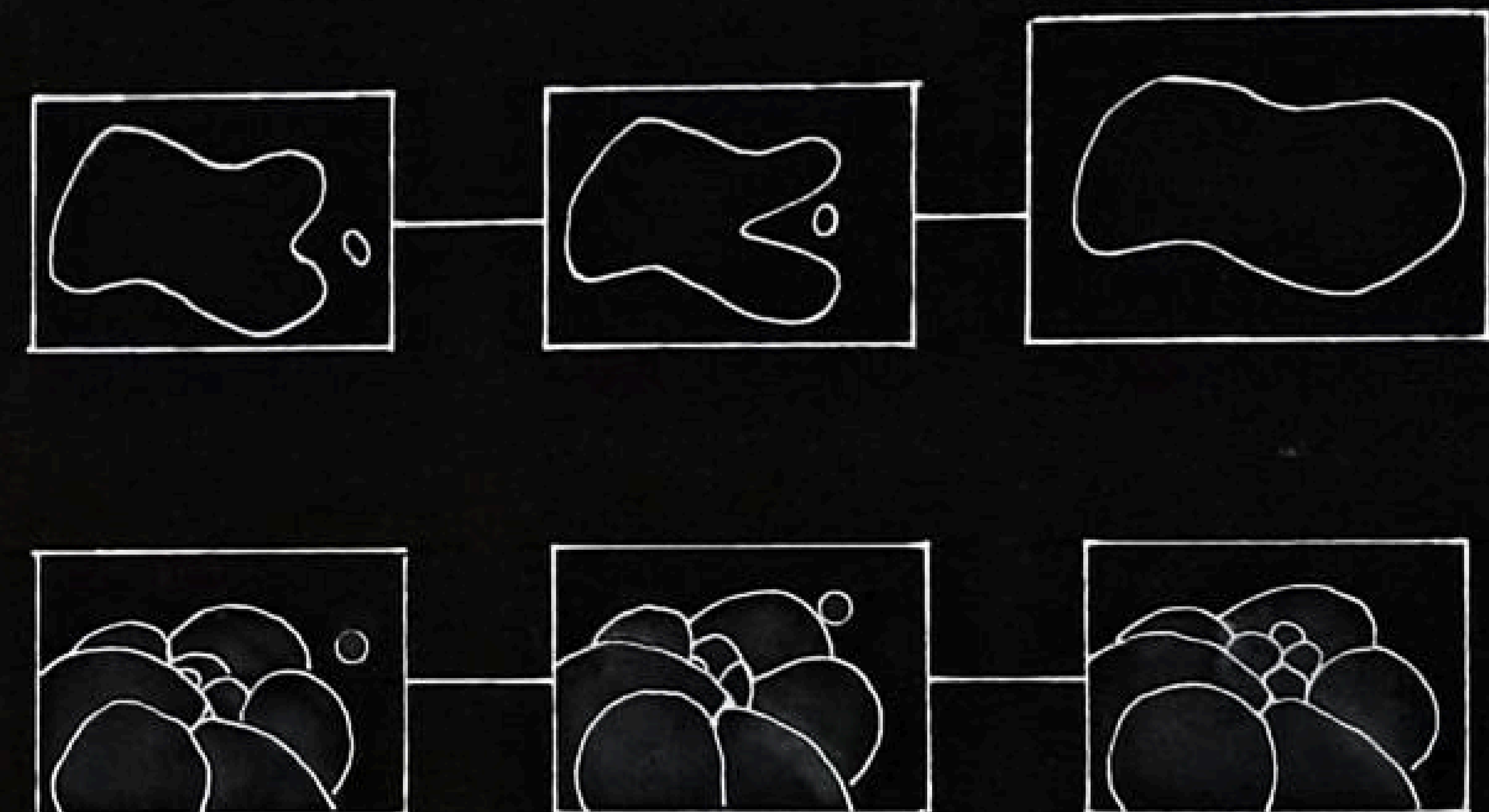


CONCEPT

AMOEBA → ABSORBS NUTRIENTS → GROWS LARGER

BUBBLE HOUSE → ABSORBS OXYGEN BUBBLES → GROWS LARGER

THIS CONCEPT TRANSFORMS ARCHITECTURE INTO A LIVING SYSTEM RATHER THAN A FIXED STRUCTURE LIKE BIOLOGICAL CELLS, BUBBLE HOUSES GROW, MERGE, DIVIDE, AND ADAPT ACCORDING TO THE NEEDS OF THEIR OCCUPANTS. THE RESULT IS A DYNAMIC AQUATIC ARCHITECTURE SHAPED BY SURVIVAL RESOURCE COLLECTION AND CONTINUOUS GROWTH.

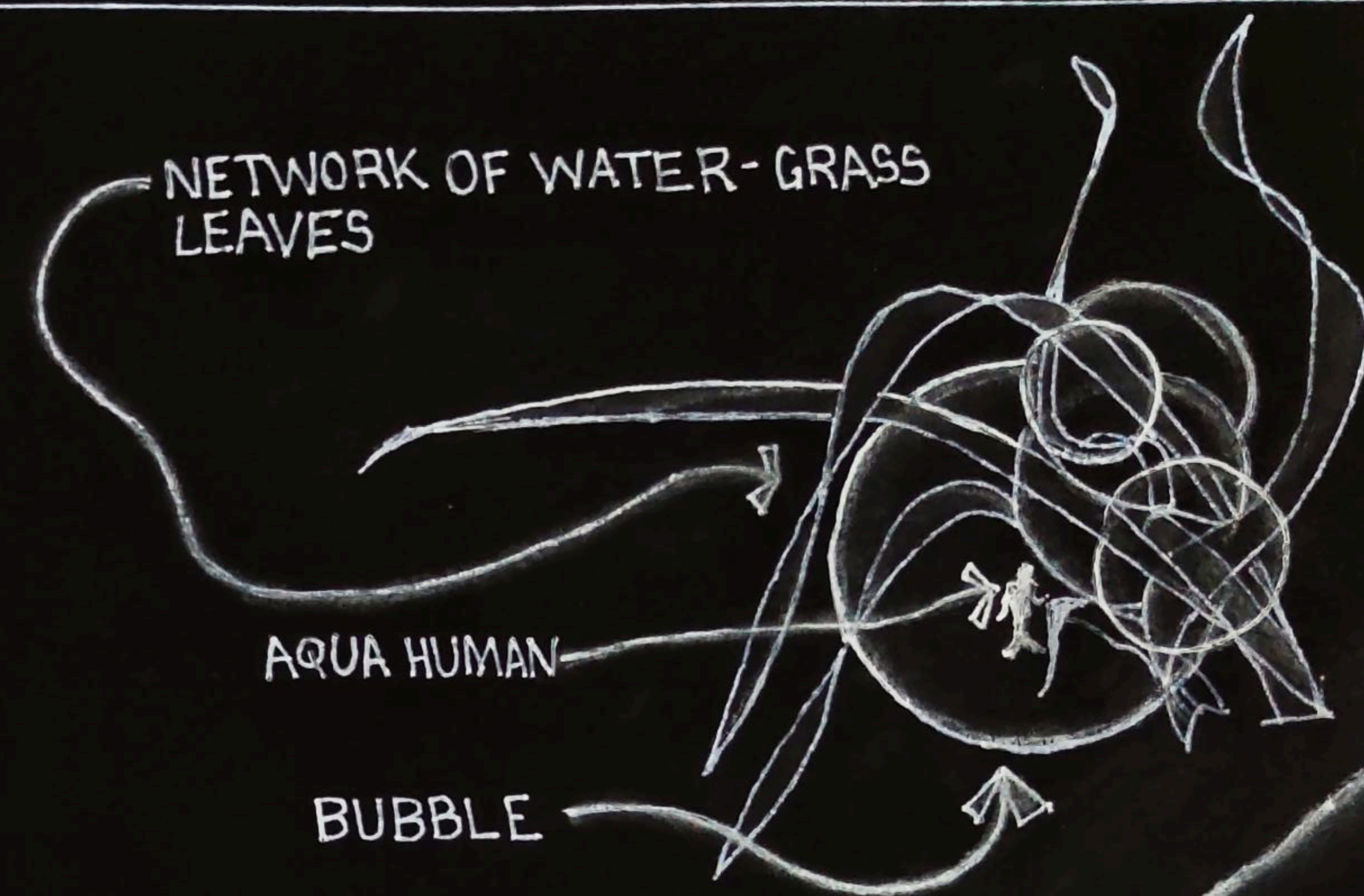


OUR DESIGN

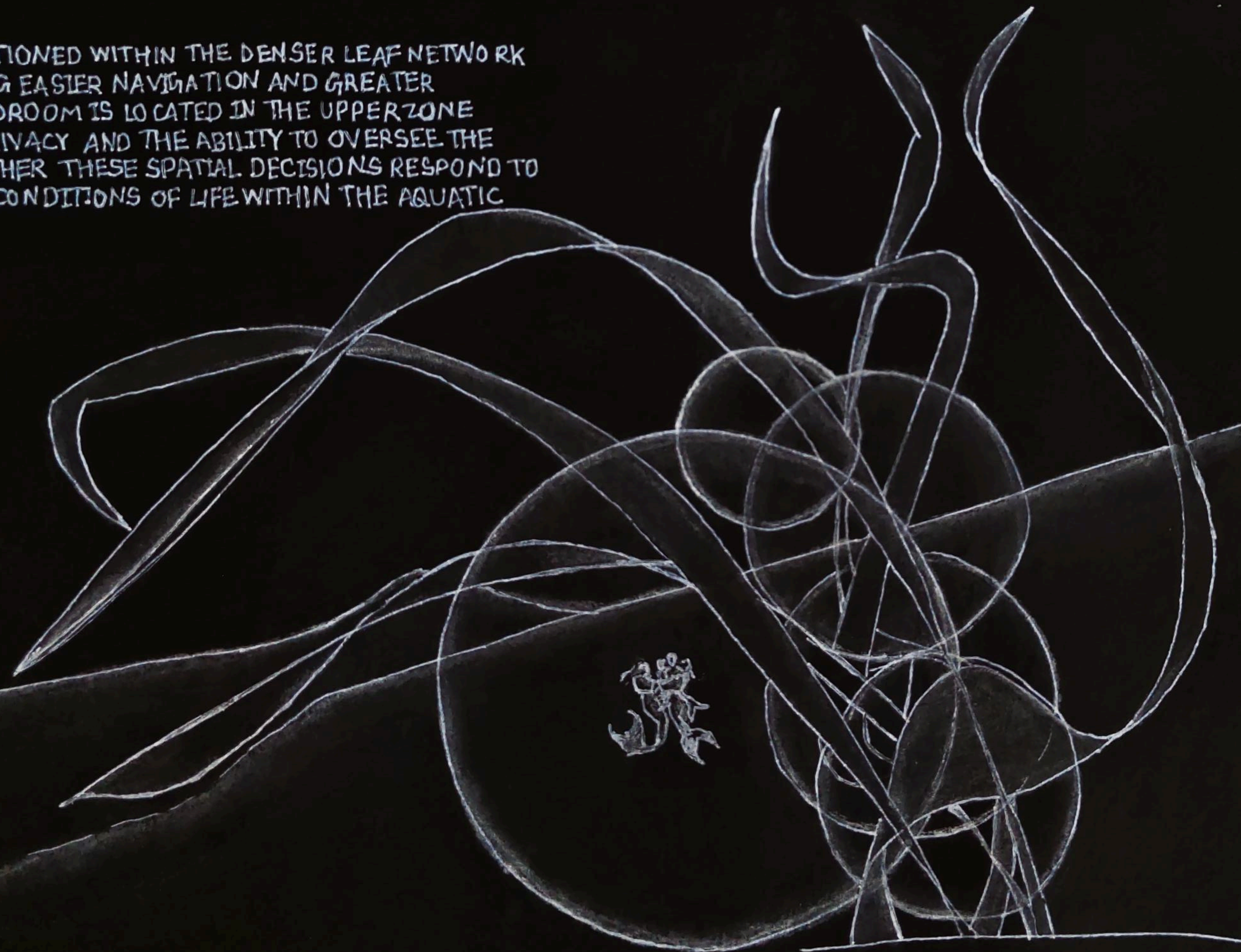
THE BUBBLE HOUSE OF MR AND MRS. VANCE IS DESIGNED FOR A FAMILY OF FOUR THE HABITAT IS DYNAMIC AND CAPABLE OF GROWTH AND ADAPTATION YET IT'S SPATIAL ORGANIZATION FOLLOWS A RADIAL ARRANGEMENT AROUND A CENTRAL NETWORK OF WATER-GRASS LEAVES. THESE LEAVES FORM A TENSILE NAVIGATION SYSTEM THAT MAINTAINS PROPORTIONAL GAPS THROUGHOUT THE CAVITY, ALLOWING SAFE MOVEMENT AND PREVENTING INHABITANTS FROM BECOMING TRAPPED WITHIN ISOLATED SPACES THE NETWORK ALSO CONTRIBUTES TO THE STRUCTURAL STABILITY OF THE HABITAT

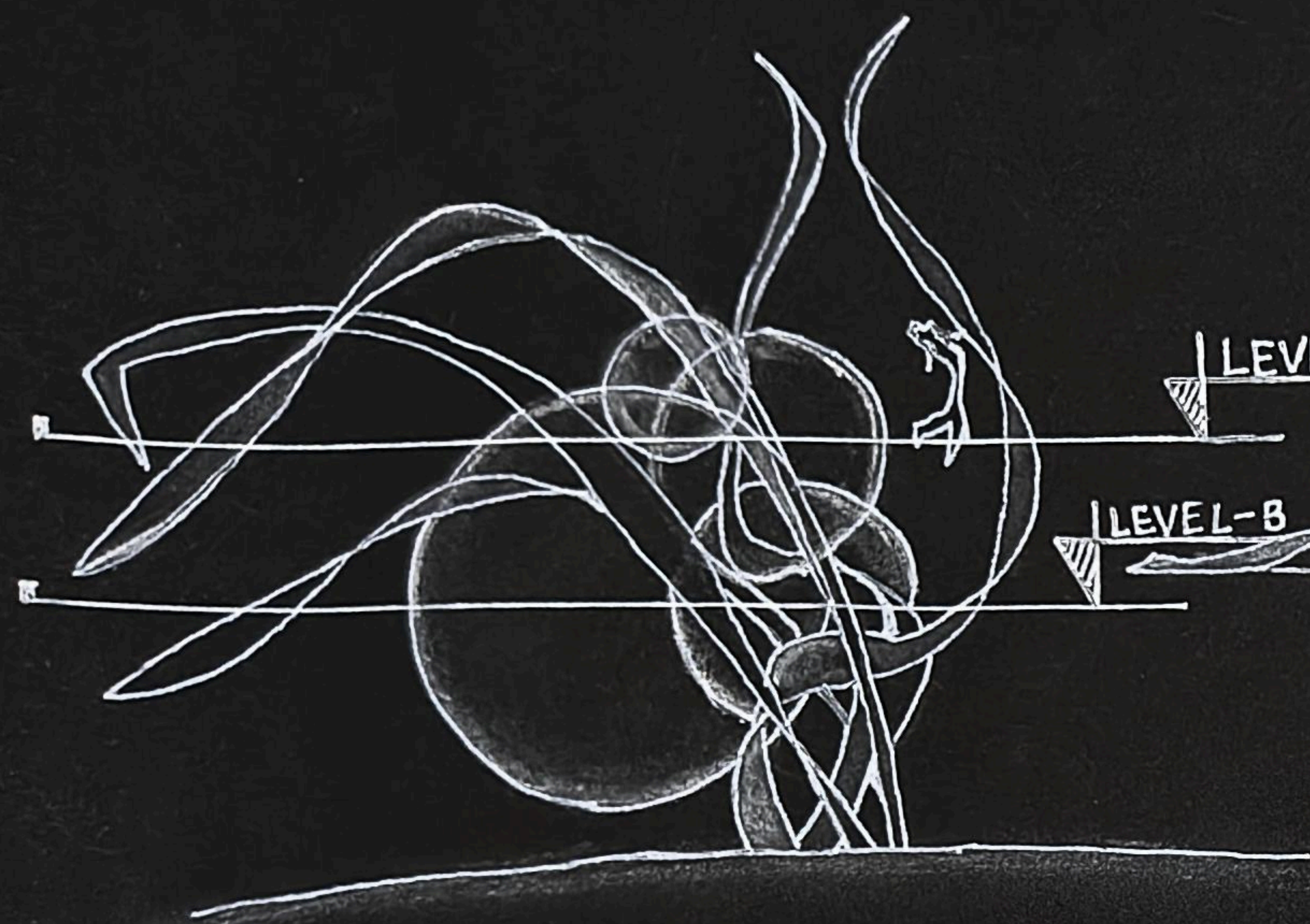
THE OXYGEN GARDEN SURROUNDS THE DWELLING AND SERVES AS BOTH A PRODUCTIVE LANDSCAPE AND A LIFE-SUPPORT SYSTEM. TOILET AND WASTE-PROCESSING AREAS ARE LOCATED WITHIN THIS GARDEN WHERE BIOLOGICAL PROCESSES CONTRIBUTE TO OXYGEN CULTIVATION AND NUTRIENT RECYCLING

THE CHILDREN'S ROOM IS POSITIONED WITHIN THE DENSER LEAF NETWORK IN THE LOWER ZONE, PROVIDING EASIER NAVIGATION AND GREATER PROTECTION THE PARENTS' BEDROOM IS LOCATED IN THE UPPER ZONE OFFERING WIDER VISIBILITY, PRIVACY AND THE ABILITY TO OVERSEE THE SURROUNDING HABITAT TOGETHER THESE SPATIAL DECISIONS RESPOND TO THE UNIQUE ENVIRONMENTAL CONDITIONS OF LIFE WITHIN THE AQUATIC WORLD.



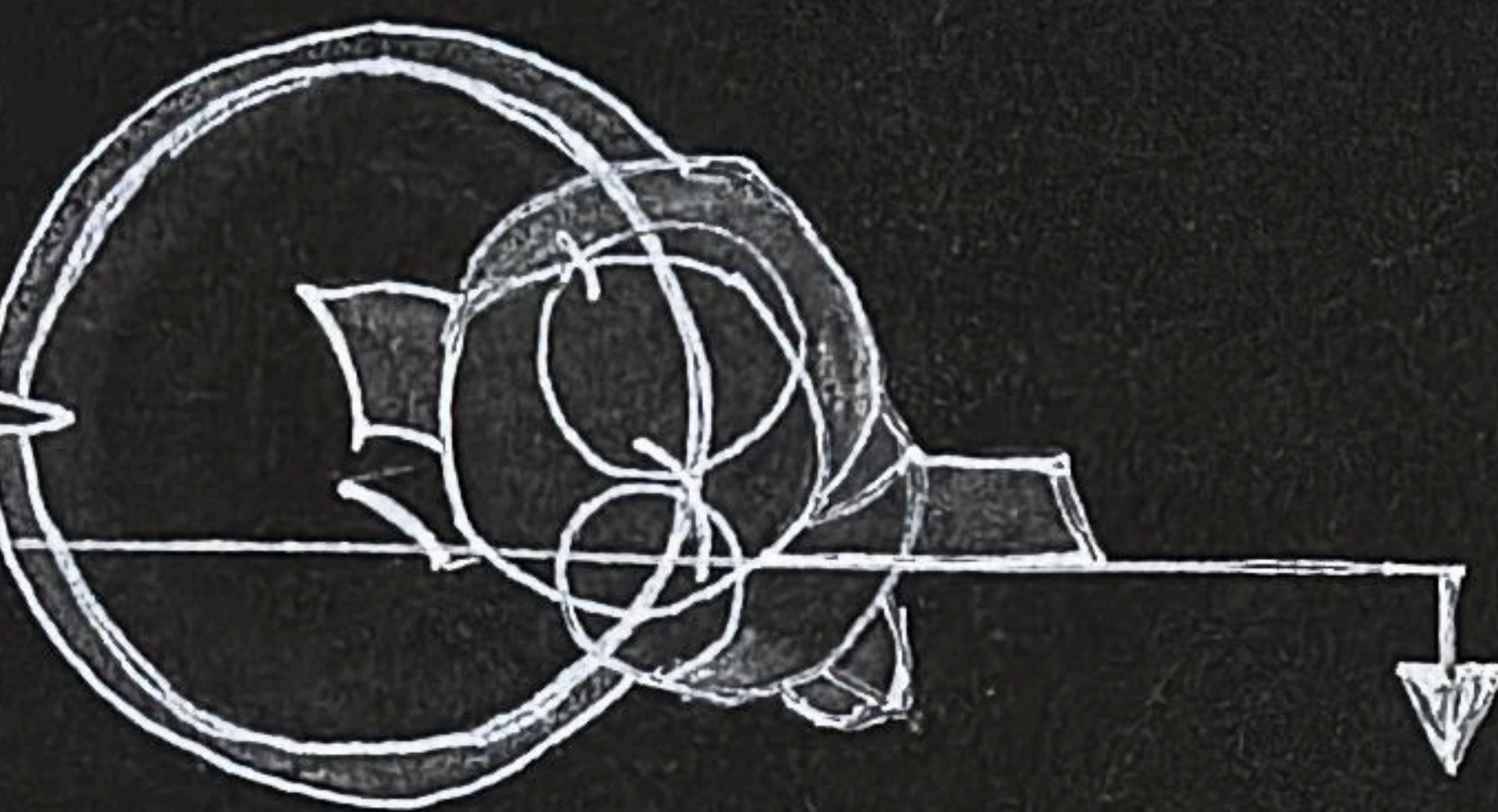
ISOMETRIC VIEW
THE BUBBLE HOUSE.



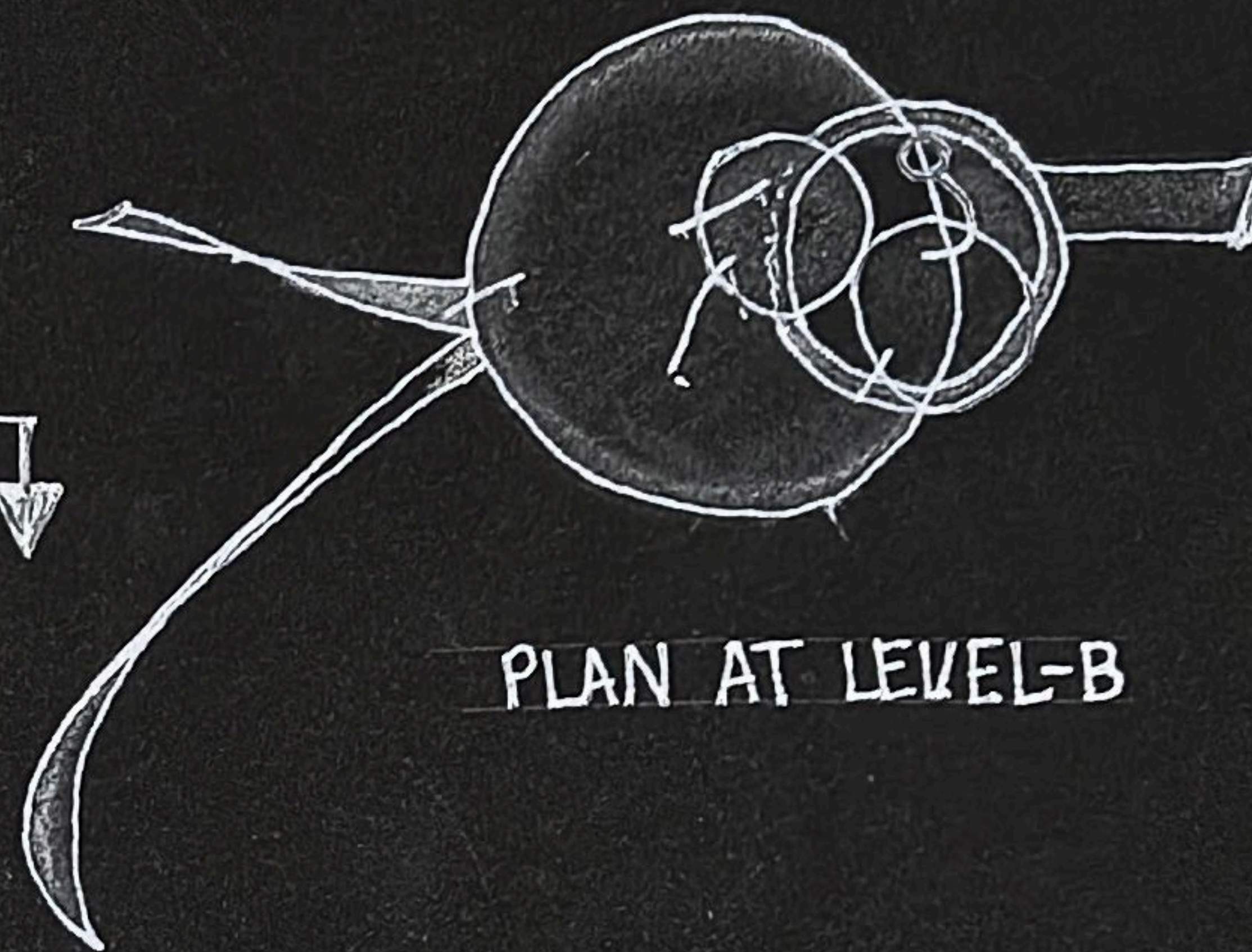


FRONT ELEVATION

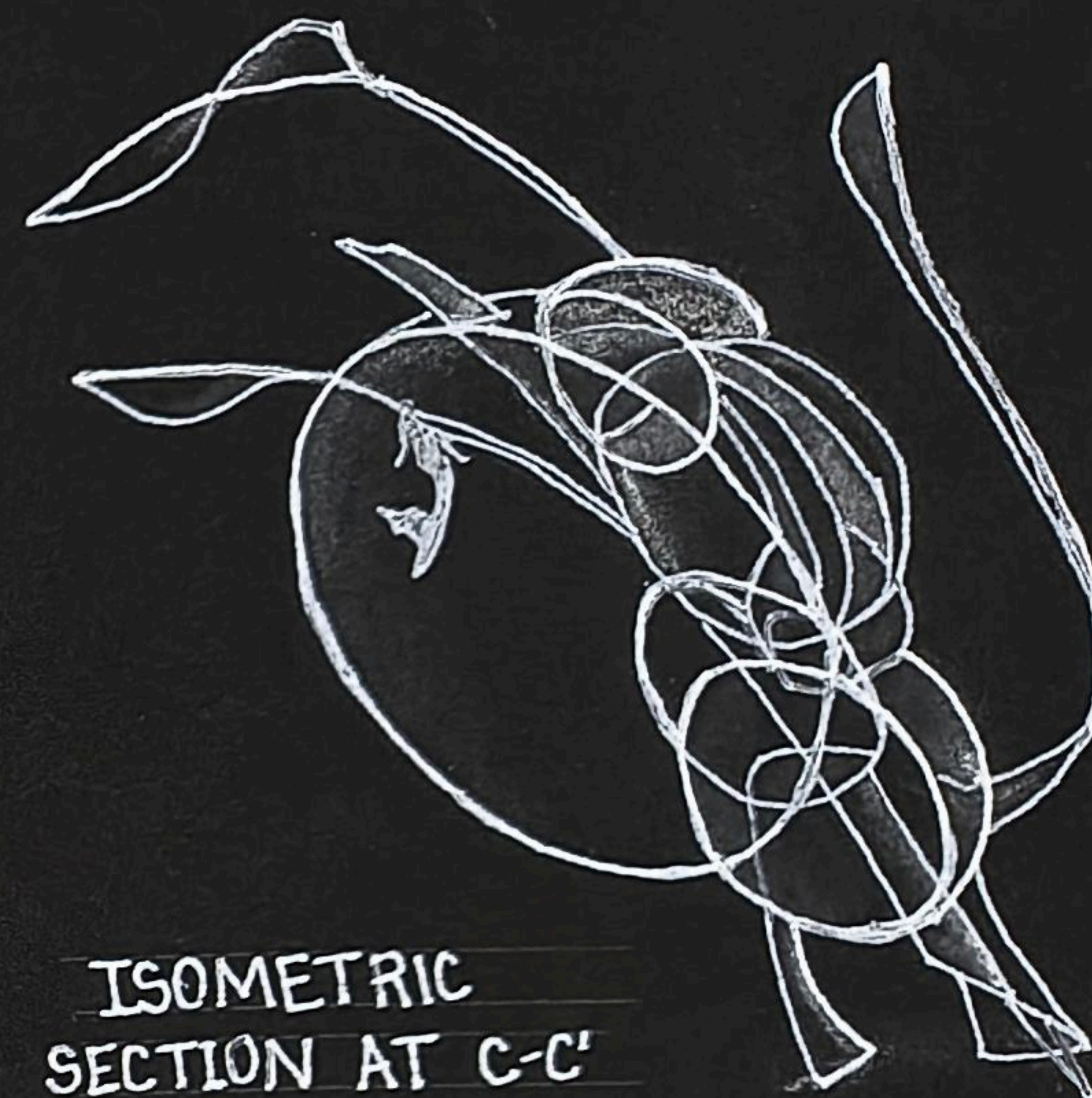
LEVEL-A
LEVEL-B



PLAN AT LEVEL-A



PLAN AT LEVEL-B



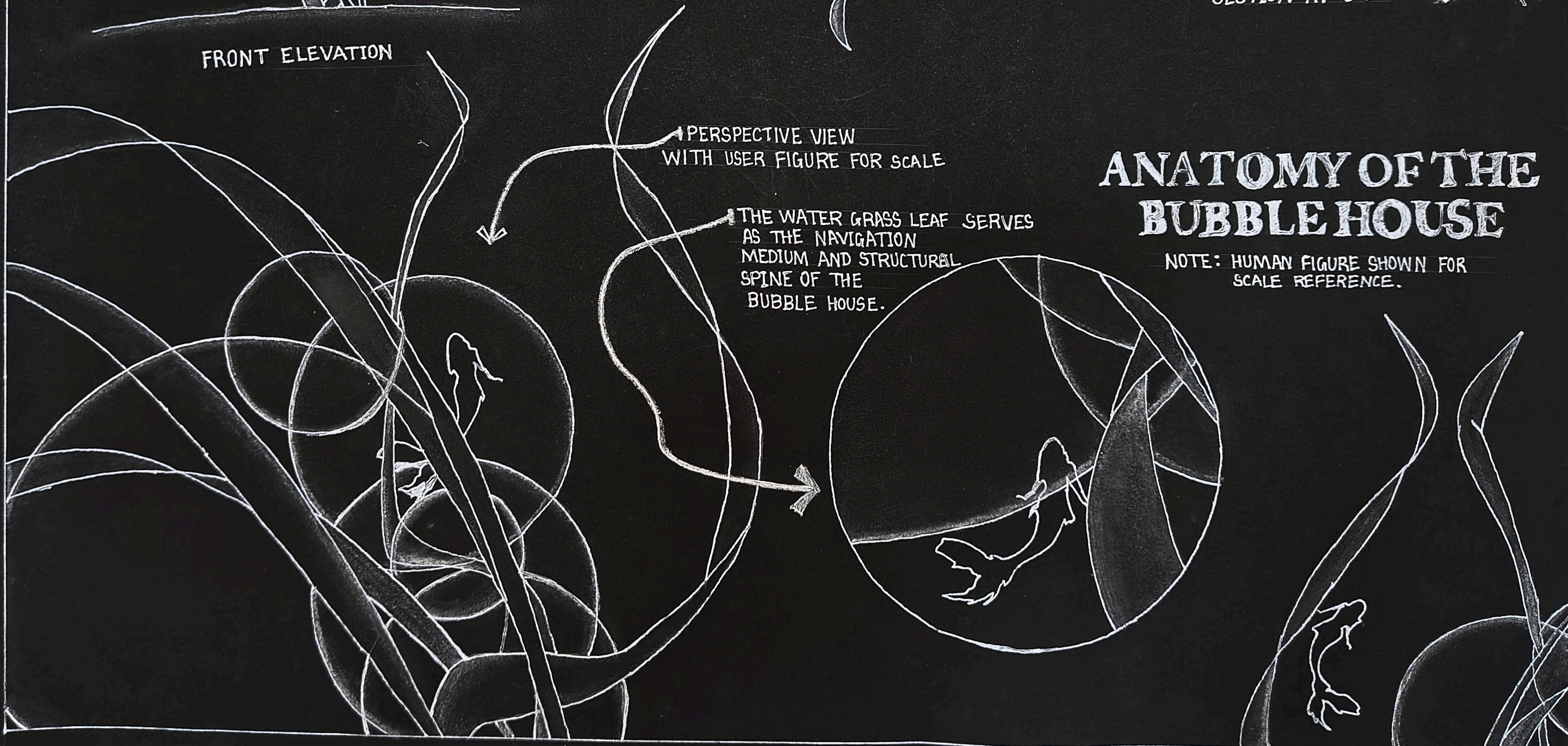
ISOMETRIC
SECTION AT C-C'

PERSPECTIVE VIEW
WITH USER FIGURE FOR SCALE

THE WATER GRASS LEAF SERVES
AS THE NAVIGATION
MEDIUM AND STRUCTURAL
SPINE OF THE
BUBBLE HOUSE.

ANATOMY OF THE BUBBLE HOUSE

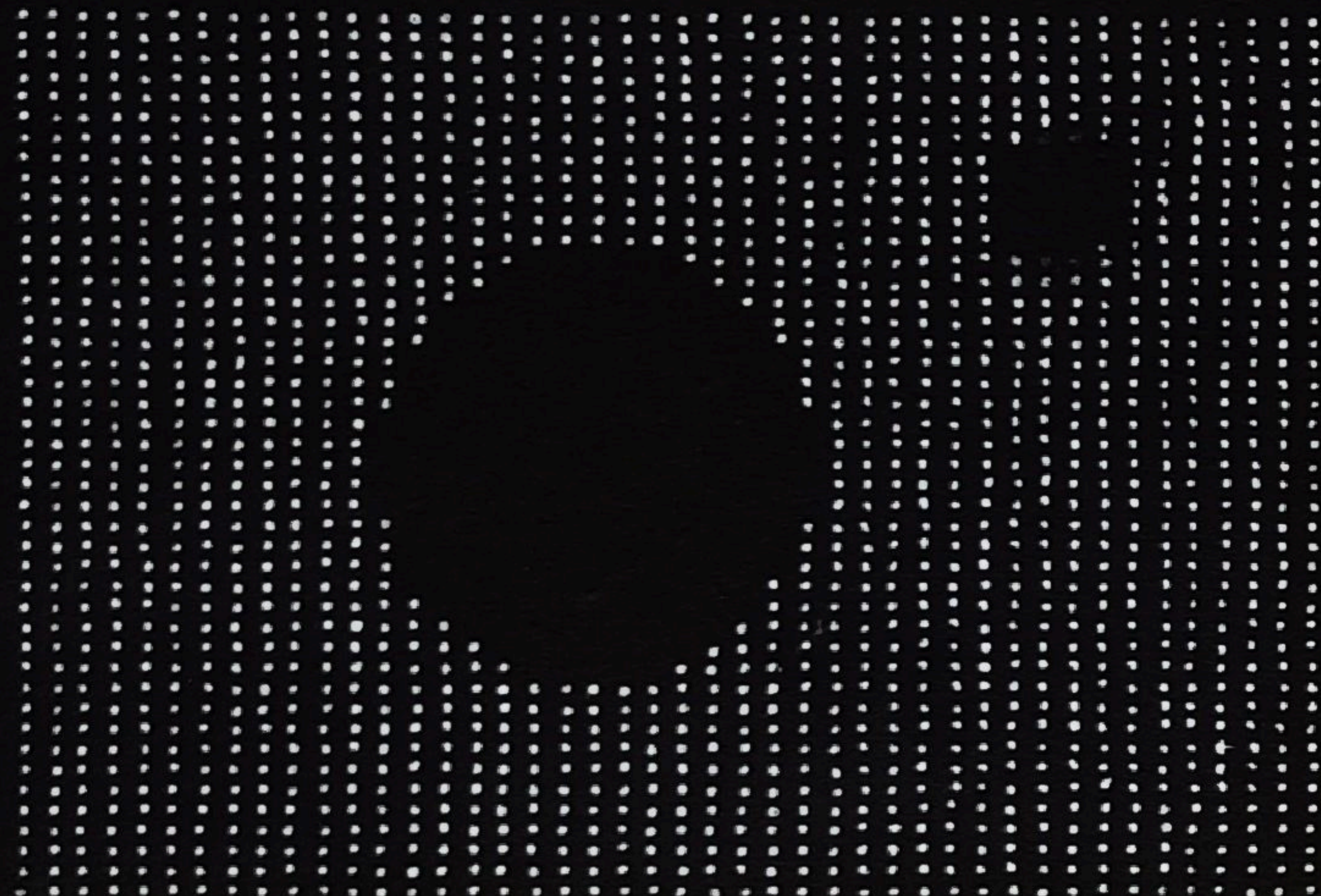
NOTE: HUMAN FIGURE SHOWN FOR
SCALE REFERENCE.



SPATIAL ANALYSIS

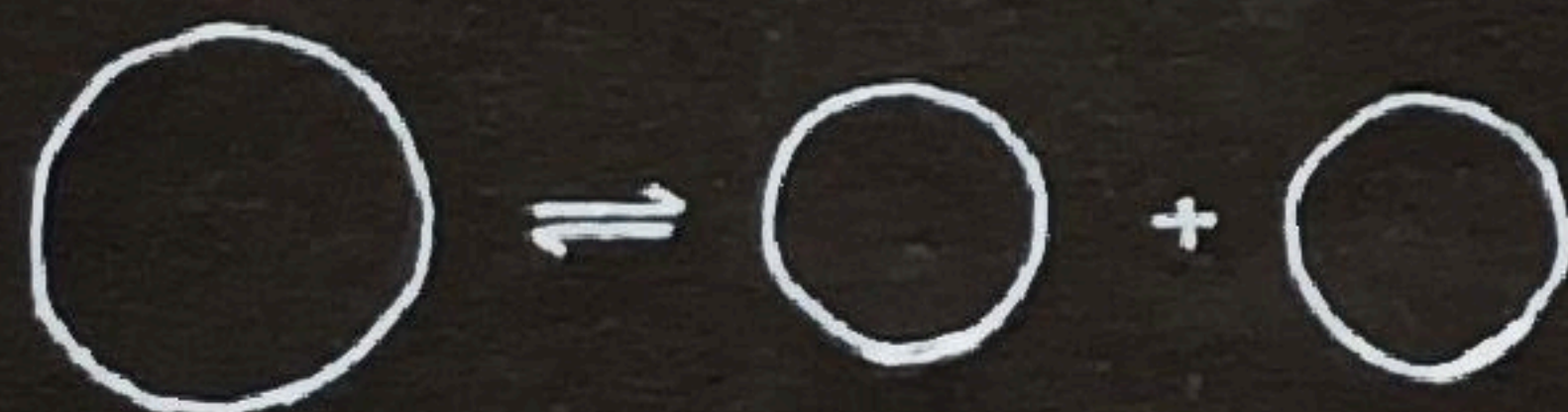
STRUCTURAL STABILITY

IN THE NEAR-ZERO-GRAVITY ENVIRONMENT OF THE DROPLET, OXYGEN CAVITIES EXPERIENCE MINIMAL BUOYANCY-DRIVEN MOVEMENT. THE BALANCE BETWEEN INTERNAL AIR PRESSURE AND SURROUNDING WATER PRESSURE MAINTAINS BUBBLE STABILITY.



EXPANSION AND DIVISION

INSPIRED BY THE BEHAVIOR OF AMOEBAS, BUBBLE HOUSES CAN GROW, MERGE, AND DIVIDE ACCORDING TO THE NEEDS OF THEIR INHABITANTS. AS SMALLER OXYGEN BUBBLES ARE CAPTURED, THEY COMBINE WHILE EXISTING HABITATS, INCREASING THEIR SIZE AND CAPACITY. WHEN A BUBBLE BECOMES TOO LARGE OR A NEW FAMILY IS FORMED, IT CAN DIVIDE INTO MULTIPLE SMALLER BUBBLES. THIS ALLOWS THE SETTLEMENT TO CONTINUOUSLY ADAPT, EXPAND, AND EVOLVE OVER GENERATIONS.

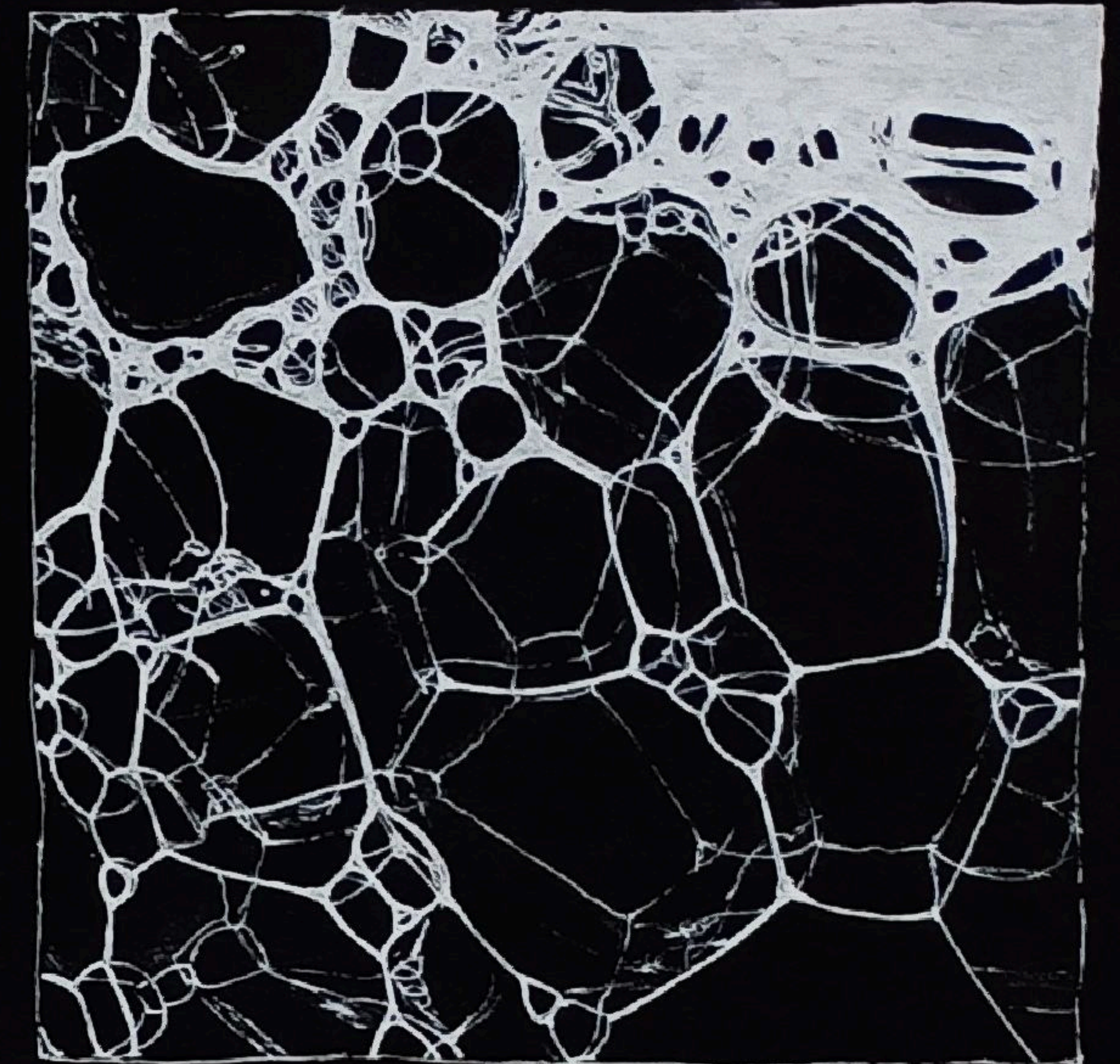


ARCHITECTURAL TRANSLATION

THE PROJECT TAKES THE FORM OF A SPOON CARRYING A SINGLE DROP OF WATER. THE SPOON SYMBOLISES HUMANITY'S EFFORT TO PRESERVE AND TRANSPORT LIFE ACROSS THE UNIVERSE, WHILE THE DROP HIGHLIGHTS HOW LITTLE WATER IS TRULY AVAILABLE AND HOW PRECIOUS IT IS FOR THE SURVIVAL OF ALL LIFE EVEN IN AN ULTRA-ADVANCED FUTURE, WATER REMAINS THE MOST FUNDAMENTAL RESOURCE WORTH CARRYING.

WITH ITS PHYSICAL FORM, THE PROJECT EXPLORES TIMELESS QUESTIONS OF EXISTENCE, PURPOSE, AND THE IDEA OF GOD. AS GENERATIONS PASS, TECHNOLOGY BECOMES MYTHOLOGY, AND THE CARETAKERS OF THE SPACECRAFT BECOME DIVINE FIGURES IN THE COLLECTIVE MEMORY OF THE INHABITANTS. THE DESIGN REFLECTS ON THE FUNDAMENTALS OF LIVING, THE MEANING OF HOME, AND HUMANITY'S RELATIONSHIP WITH LIFE ITSELF.

INSPIRED BY THE ADAPTIVE BEHAVIOR OF AMOEBAS WHICH IS ALSO THE OLDEST LIVING CELL, THE BUBBLE HOUSES GROW, MERGE, AND DIVIDE OVER TIME, ALLOWING THE SETTLEMENT TO EVOLVE WITH ITS INHABITANTS THROUGH THE SPECULATIVE NARRATIVE, THE PROJECT ATTEMPTS TO THINK BEYOND CONVENTIONAL ARCHITECTURE AND IMAGINE A FUTURE WHERE A SINGLE DROP OF WATER BECOMES AN ENTIRE WORLD.



LIQUID FRESH
WATER

FRESH-WATER
LAKES AND RIVER

ALL WATER ON, IN,
AND ABOVE
THE EARTH

SCALE COMPARISON

